

Deepak Mallampati

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PROFESSIONAL SUMMARY

Healthcare AI engineer with 2.5+ years building clinical data pipelines, RAG systems for medical knowledge retrieval, predictive ML models for patient care, and AI governance frameworks in HIPAA-conscious environments. Proven ability to design ETL workflows for EHR data, validate clinical code logic (CPT/ICD), and deliver AI-assisted solutions that improve care outcomes. Strong Python, SQL, and FastAPI foundation with deep expertise in responsible AI development, audit-ready documentation, and technical advising for healthcare stakeholders. Master's degree in Computer Science.

CORE COMPETENCIES

Healthcare AI • ETL Pipeline Development • Clinical Data Integration • CPT/ICD Validation • EHR Data Processing • RAG Systems • Agentic AI Workflows • Python • SQL (T-SQL, PostgreSQL) • FastAPI • Docker • Data Quality Validation • HIPAA Compliance • AI Governance • LLM Evaluation • scikit-learn • XGBoost

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer - Healthcare Technology & AI Consulting

- Built **clinical data ETL pipelines** (Pandas, NumPy, SQL) processing EHR extracts, appointment histories, and support ticket logs; raised dataset reliability above **98%** and cut downstream model instability by **~40%**; implemented de-identification and data lineage documentation for HIPAA compliance.
- Built **RAG systems** for clinical knowledge retrieval and healthcare documentation search; recursive semantic chunking and transformer-based embeddings improved contextual retrieval precision by **~35%**; response alignment exceeded **90%** in evaluation.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction and care engagement scoring; improved recall by **15-20%** on high-risk categories; class weighting, stratified sampling, and threshold calibration kept precision **>90%**.
- Developed **agentic LLM workflows** for eligibility checks, care workflow navigation, and documentation clarification with structured reasoning and grounding rules; improved response stability by **~25%**.
- Maintained **HIPAA-conscious AI governance**: evaluation audit trails, system-limitation documentation, and stakeholder-facing compliance reports - directly supporting safe AI deployment at health systems.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services containerized with **Docker**; structured logging and load simulation reduced post-deployment defects by **~30%**.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Enterprise ERP

- Optimized **T-SQL stored procedures** for an Order-to-Cash platform; reduced query execution time **~20%** and data retrieval latency **~25%**.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; reduced deployment errors **>30%**.
- Designed performance dashboards (SQL Server DMVs + **Grafana**) tracking long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence **~25%**.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Applied NLP

- Implemented **document Q&A with RAG**; reduced hallucinated outputs **~30%** and raised contextual accuracy **>85%**.
- Deployed as **Flask-based API** with a web interface for non-technical staff.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents**. **RAG-grounded CPT/ICD validation** against vector-indexed policy documents; ANN similarity search for duplicate claim detection; explainable risk scoring with audit-ready reasoning traces - directly applicable to healthcare AI solutions engineering.

Python, LangChain, LlamaIndex, RAG, CPT/ICD validation, multi-agent orchestration, vector search, FastAPI

Patient Records Management Web Application

Bottle + SQLite relational healthcare CRUD app with normalized **patients/diagnoses** schema, foreign-key constraints, and **LEFT JOIN + GROUP_CONCAT** aggregation for single-query patient+diagnosis retrieval.

Python, SQLite, SQL, REST APIs, healthcare data modeling

IoT + ML Smart Building Temperature Prediction

Time-series forecasting pipeline with walk-forward validation; lag-window feature engineering ($n_{in}=6$); Random Forest achieved **~20-30% lower MAE** vs. linear baseline.

Python, scikit-learn, Pandas, time-series forecasting, feature engineering

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Healthcare & Data: EHR data processing, CPT/ICD validation, clinical data pipelines, ETL, data quality validation, HIPAA compliance, data lineage documentation

AI & ML: RAG, agentic AI workflows, LLM evaluation, LangChain, LlamaIndex, scikit-learn, XGBoost, PyTorch, Hugging Face Transformers

Languages & Frameworks: Python, SQL (T-SQL, PostgreSQL, SQLite), FastAPI, Flask, Pandas, NumPy

Infra: Docker, Jenkins, Grafana, CI/CD, RESTful APIs, cloud-ready deployment, observability/logging

Work Authorization: F-1 OPT - authorized to work in the United States now